

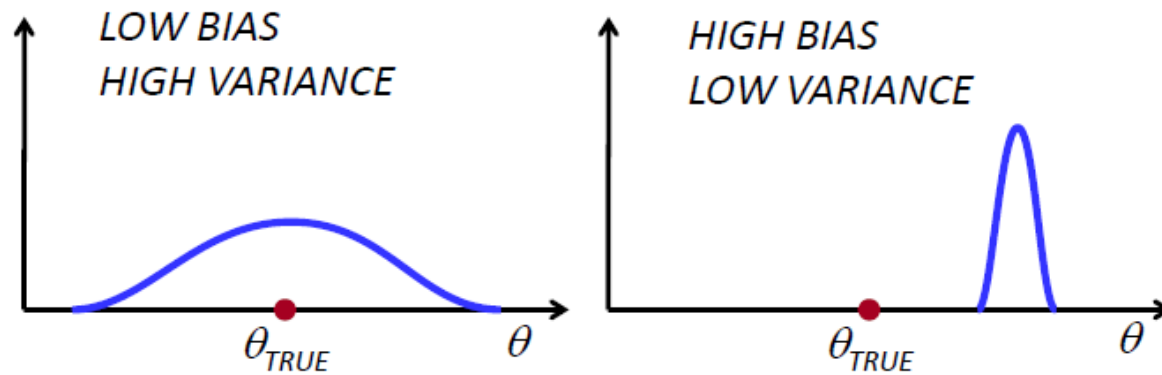
# *Biased and Unbiased* *Estimators*

# Biased and Unbiased Estimators

- An estimator of a parameter is **biased** if the expected value of the estimate is **different from** the true value of the parameters.
- An estimator of a parameter is **unbiased** if the expected value of the estimate is the **same** as the true value of the parameters.

# Bias and Variance

- The **bias-variance tradeoff**: in most cases, you can only decrease one of them at the expense of the other



# ***Biased and Unbiased Estimates***

- An estimate  $\hat{\theta}$  is **unbiased** when

$$E[\hat{\theta}] = \theta$$

- The ML estimate  $\hat{\mu}$  is **unbiased**, i.e.,

$$E[\hat{\mu}] = \mu$$

- The ML estimates  $\hat{\sigma}$  and  $\hat{\Sigma}$  are **biased**:

$$E[\hat{\sigma}^2] = \frac{n-1}{n} \sigma^2 \quad E[\hat{\Sigma}] = \frac{n-1}{n} \Sigma$$

## Gaussian Case: Unknown Mean

- Consider the case where only the mean,  $\theta = \mu$ , is unknown:

$$\sum_{k=1}^n \nabla_{\theta} \ln(p(\mathbf{x}_k | \theta)) = 0$$

$$\begin{aligned} \ln(p(\mathbf{x}_k | \theta)) &= \ln\left[\frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{x}_k - \theta)^t \Sigma^{-1}(\mathbf{x}_k - \theta)\right]\right] \\ &= -\frac{1}{2} \ln[(2\pi)^d |\Sigma|] - \frac{1}{2}(\mathbf{x}_k - \theta)^t \Sigma^{-1}(\mathbf{x}_k - \theta) \end{aligned}$$

**which implies:**  $\nabla_{\theta} \ln(p(\mathbf{x}_k | \theta)) = \Sigma^{-1}(\mathbf{x}_k - \theta)$

**because:**

$$\begin{aligned} &\frac{\partial}{\partial \theta} \left\{ \left[ -\frac{1}{2} \ln[(2\pi)^d |\Sigma|] - \frac{1}{2}(\mathbf{x}_k - \theta)^t \Sigma^{-1}(\mathbf{x}_k - \theta) \right] \right\} \\ &= \frac{\partial}{\partial \theta} \left[ -\frac{1}{2} \ln[(2\pi)^d |\Sigma|] \right] - \frac{\partial}{\partial \theta} \left[ \frac{1}{2}(\mathbf{x}_k - \theta)^t \Sigma^{-1}(\mathbf{x}_k - \theta) \right] \\ &= \Sigma^{-1}(\mathbf{x}_k - \theta) \end{aligned}$$

## Gaussian Case: Unknown Mean

- Substituting into the expression for the total likelihood:

$$\nabla_{\theta} l = \sum_{k=1}^n \nabla_{\theta} \ln(p(\mathbf{x}_k | \theta)) = \sum_{k=1}^n \Sigma^{-1} (\mathbf{x}_k - \theta) = 0$$

- Rearranging terms:  $\sum_{k=1}^n \Sigma^{-1} (\mathbf{x}_k - \hat{\theta}) = 0$

$$\sum_{k=1}^n (\mathbf{x}_k - \hat{\theta}) = 0$$

$$\sum_{k=1}^n \mathbf{x}_k - \sum_{k=1}^n \hat{\theta} = 0$$

$$\sum_{k=1}^n \mathbf{x}_k - n \hat{\theta} = 0$$

$$\hat{\theta} = \frac{1}{n} \sum_{k=1}^n \mathbf{x}_k$$

## Gaussian Case: Unknown Mean and Variance

- Let  $\theta = [\mu, \sigma^2]$ . The log likelihood of a SINGLE point is:

$$\ln(p(x_k|\theta)) = -\frac{1}{2} \ln[(2\pi)\theta_2] - \frac{1}{2} (x_k - \theta_1)^t \theta_2^{-1} (x_k - \theta_1)$$

$$\nabla_{\theta} l = \nabla_{\theta} \ln(p(x_k|\theta)) = \begin{bmatrix} \frac{1}{\theta_2} (x_k - \theta_1) \\ -\frac{1}{2\theta_2} + \frac{(x_k - \theta_1)^2}{2\theta_2^2} \end{bmatrix}$$

- The full likelihood leads to:

$$\sum_{k=1}^n \frac{1}{\hat{\theta}_2} (x_k - \hat{\theta}_1) = 0$$

$$\sum_{k=1}^n -\frac{1}{2\hat{\theta}_2} + \frac{(x_k - \hat{\theta}_1)^2}{2\hat{\theta}_2^2} = 0 \Rightarrow \sum_{k=1}^n (x_k - \hat{\theta}_1)^2 = \sum_{k=1}^n \hat{\theta}_2$$

## Gaussian Case: Unknown Mean and Variance

- This leads to these equations:  $\hat{\theta}_1 = \hat{\mu} = \frac{1}{n} \sum_{k=1}^n x_k$

$$\hat{\theta}_2 = \hat{\sigma}^2 = \frac{1}{n} \sum_{k=1}^n (x_k - \hat{\mu})^2$$

- In the multivariate case:  $\hat{\mu} = \frac{1}{n} \sum_{k=1}^n \mathbf{x}_k$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{k=1}^n (\mathbf{x}_k - \hat{\mu})(\mathbf{x}_k - \hat{\mu})^t$$

- The true covariance is the expected value of the matrix  $(\mathbf{x}_k - \hat{\mu})(\mathbf{x}_k - \hat{\mu})^t$ , which is a familiar result.

# Convergence of the Mean

- Does the maximum likelihood estimate of the variance converge to the true value of the variance? Let's start with a few simple results we will need later.
- Expected value of the ML estimate of the mean:

$$\begin{aligned}E[\hat{\mu}] &= E\left[\frac{1}{n} \sum_{i=1}^n x_i\right] \\&= \frac{1}{n} \sum_{i=1}^n E[x_i] \\&= \frac{1}{n} \sum_{i=1}^n \mu = \mu\end{aligned}$$

$$\begin{aligned}\text{var}[\hat{\mu}] &= E[\hat{\mu}^2] - (E[\hat{\mu}])^2 \\&= E[\hat{\mu}^2] - \mu^2 \\&= E\left[\left(\frac{1}{n} \sum_{i=1}^n x_i\right)\left(\frac{1}{n} \sum_{j=1}^n x_j\right)\right] - \mu^2 \\&= \frac{1}{n^2} \left( \sum_{i=1}^n \sum_{j=1}^n E[x_i x_j] \right) - \mu^2\end{aligned}$$

## Variance of the ML Estimate of the Mean

- The expected value of  $x_i x_j$ ,  $E[x_i x_j]$ , will be  $\mu^2$  for  $i \neq j$  and  $\mu^2 + \sigma^2$  otherwise since the two random variables are independent.
- The expected value of  $x_i^2$  will be  $\mu^2 + \sigma^2$ .
- Hence, in the summation above, we have  $n^2 - n$  terms with expected value  $\mu^2$  and  $n$  terms with expected value  $\mu^2 + \sigma^2$ .
- Thus,

$$\text{var}[\hat{\mu}] = \frac{1}{n^2} \left( (n^2 - n)\mu^2 + n(\mu^2 + \sigma^2) \right) - \mu^2 = \frac{\sigma^2}{n}$$

which implies:

$$E[\hat{\mu}^2] = \text{var}[\hat{\mu}] + (E[\hat{\mu}])^2 = \frac{\sigma^2}{n} + \mu^2$$

- We see that the variance of the estimate goes to zero as  $n$  goes to infinity, and our estimate converges to the true estimate (error goes to zero).

# Variance Relationships

- We will need one more result:

$$\begin{aligned}\sigma^2 &= E[(x - \mu)^2] = E[x^2] - 2E[x]\mu + E[\mu^2] \\ &= E[x^2] - 2\mu^2 + E[\mu^2] \\ &= E[x^2] - \mu^2 \\ &= \sum_{i=1}^n x_i^2 - \left(\frac{1}{n} \sum_{i=1}^n x_i\right)^2\end{aligned}$$

**Note that this implies:**

$$\sum_{i=1}^n x_i^2 = \sigma^2 + \mu^2$$

- Now we can combine these results. Recall our expression for the ML estimate of the variance:

$$\hat{\sigma}^2 = E\left[\frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2\right]$$

# Covariance Expansion

- Expand the covariance and simplify:

$$\begin{aligned}\hat{\sigma}^2 &= E\left[\frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2\right] = \frac{1}{n} E\left[\sum_{i=1}^n (x_i^2 - 2x_i\hat{\mu} + \hat{\mu}^2)\right] \\ &= \frac{1}{n} \sum_{i=1}^n (E[x_i^2] - 2E[x_i\hat{\mu}] + E[\hat{\mu}^2]) \\ &= \frac{1}{n} \sum_{i=1}^n ((\sigma^2 + \mu^2) - 2E[x_i\hat{\mu}] + (\mu^2 + \sigma^2/n))\end{aligned}$$

- One more intermediate term to derive:

$$\begin{aligned}E[x_i\hat{\mu}] &= E\left[x_i \frac{1}{n} \sum_{j=1}^n x_j\right] = \frac{1}{n} \sum_{j=1}^n E[x_i x_j] = \frac{1}{n} \left( \sum_{\substack{j=1 \\ i \neq j}}^n E[x_i x_j] + E[x_i x_i] \right) \\ &= \frac{1}{n} ((n-1)\mu^2 + (\mu^2 + \sigma^2)) = \frac{1}{n} (n\mu^2 + \sigma^2) = \mu^2 + \frac{\sigma^2}{n}\end{aligned}$$

# Biased Variance Estimate

- Substitute our previously derived expression for the second term:

$$\begin{aligned}\hat{\sigma}^2 &= \frac{1}{n} \sum_{i=1}^n ((\sigma^2 + \mu^2) - 2E[x_i \hat{\mu}] + (\mu^2 + \sigma^2/n)) \\&= \frac{1}{n} \sum_{i=1}^n ((\sigma^2 + \mu^2) - 2(\mu^2 + \sigma^2/n) + (\mu^2 + \sigma^2/n)) \\&= \frac{1}{n} \sum_{i=1}^n (\sigma^2 + \mu^2 - 2\mu^2 + \mu^2 - 2\sigma^2/n + \sigma^2/n) \\&= \frac{1}{n} \sum_{i=1}^n (\sigma^2 - \sigma^2/n) \\&= \frac{1}{n} \sum_{i=1}^n (\sigma^2 - \sigma^2/n) = \frac{1}{n} \sum_{i=1}^n \sigma^2 (1 - 1/n) = \frac{1}{n} \sum_{i=1}^n \sigma^2 \frac{(n-1)}{n} \\&= \frac{(n-1)}{n} \sigma^2\end{aligned}$$

# Expectation Simplification

- Therefore, the ML estimate is biased:

$$\hat{\sigma}^2 = E\left[\frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2\right] = \frac{n-1}{n} \sigma^2 \neq \sigma^2$$

However, the ML estimate converges .

- An unbiased estimator is:

$$\mathbf{C} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \hat{\mu})(\mathbf{x}_i - \hat{\mu})^t$$

III Sample Variance of ML estimator

$$S^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$\begin{aligned} \therefore E(S^2) &= E\left(\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2\right) \\ &= \frac{1}{n} E\left[\sum_{i=1}^n x_i^2 - 2x_i \bar{x} + \bar{x}^2\right] \\ &= \frac{1}{n} E\left[\sum x_i^2 - 2 \sum x_i \bar{x} + \sum \bar{x}^2\right] \\ &= \frac{1}{n} E\left[\sum x_i^2 - 2(\bar{x}n)\bar{x} + n\bar{x}^2\right] \\ &= \frac{1}{n} E\left[\sum x_i^2 - 2n\bar{x}^2 + n\bar{x}^2\right] \\ &= \frac{1}{n} E\left[\sum x_i^2 - n\bar{x}^2\right] \\ &= \frac{1}{n} \sum E(x^2) - E(\bar{x}^2) \\ &= \frac{1}{n} \times n E(x^2) - E(\bar{x}^2) \end{aligned}$$

$$\boxed{E(S^2) = E(x^2) - E(\bar{x}^2)}$$

Now,  $\sigma_{x^2} = E(\bar{x}^2) - [E(x)]^2$

or  $\sigma_{x^2} = E(x^2) - \mu^2$

$$\therefore \boxed{E(x^2) = \sigma^2 + \mu^2}$$

Now,  $\boxed{\sigma_{\bar{x}}^2 = E(\bar{x}^2) - [E(\bar{x})]^2}$

Now,  $\text{Var}(x_1 + x_2 + x_3 + \dots + x_n) = n \cdot \sigma_{x^2}$

$$\text{Var}\left(\frac{x_1 + x_2 + x_3 + \dots + x_n}{n}\right) = \frac{n \sigma_{x^2}}{n^2} = \frac{\sigma_{x^2}}{n} = \sigma_{\bar{x}}^2$$

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$$\therefore \sigma_{\bar{x}}^2 = E(\bar{x}^2) - [E(\bar{x})]^2$$

$$\frac{\sigma_{x^2}}{n} = E(\bar{x}^2) - \underbrace{[E(\bar{x})]^2}_{[E(x)]^2}$$

$$\therefore \frac{\sigma_{x^2}}{n} = E(\bar{x}^2) - [E(x)]^2$$

Here  $E(x)$  is  $\mu$

$$\therefore \frac{\sigma_{x^2}}{n} = E(\bar{x}^2) - \mu^2$$

$$\Rightarrow E(\bar{x}^2) = \frac{\sigma_{x^2}}{n} + \mu^2$$

Now, we know

$$\begin{aligned} E(s^2) &= E(x^2) - E(\bar{x}^2) \\ &= (n\sigma^2 + n\mu^2) - \left(\frac{\sigma_{x^2}}{n} + \mu^2\right) \\ &= \frac{n\sigma^2 + n\mu^2 - \sigma_{x^2} + n\mu^2}{n} \\ &= \frac{(n-1)\sigma^2}{n} \end{aligned}$$

$$\therefore E(s^2) = \left(\frac{n-1}{n}\right) \sigma^2$$